LuatCheck

A local, evidence-grounded Vietnamese legal assistant

Legal answers you can verify

D1 proposal pitch · Group 8 · Theme 9 · Application track
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Problem and users

0:00 – 0:30

Who asks

Vietnamese non-lawyers — employees, small-business owners, SME HR and administrative staff.

“I am being let go next month — how many days of notice am I entitled to?”

Two options today, both fail

- Public chatbot — the question leaves the organization, and the article it cites may not support the claim
- Legal portal — authoritative text, but you must already know which document to open
- Commercial legal AI — mis-citation still measured in deployed tools (Magesh et al. 2025)

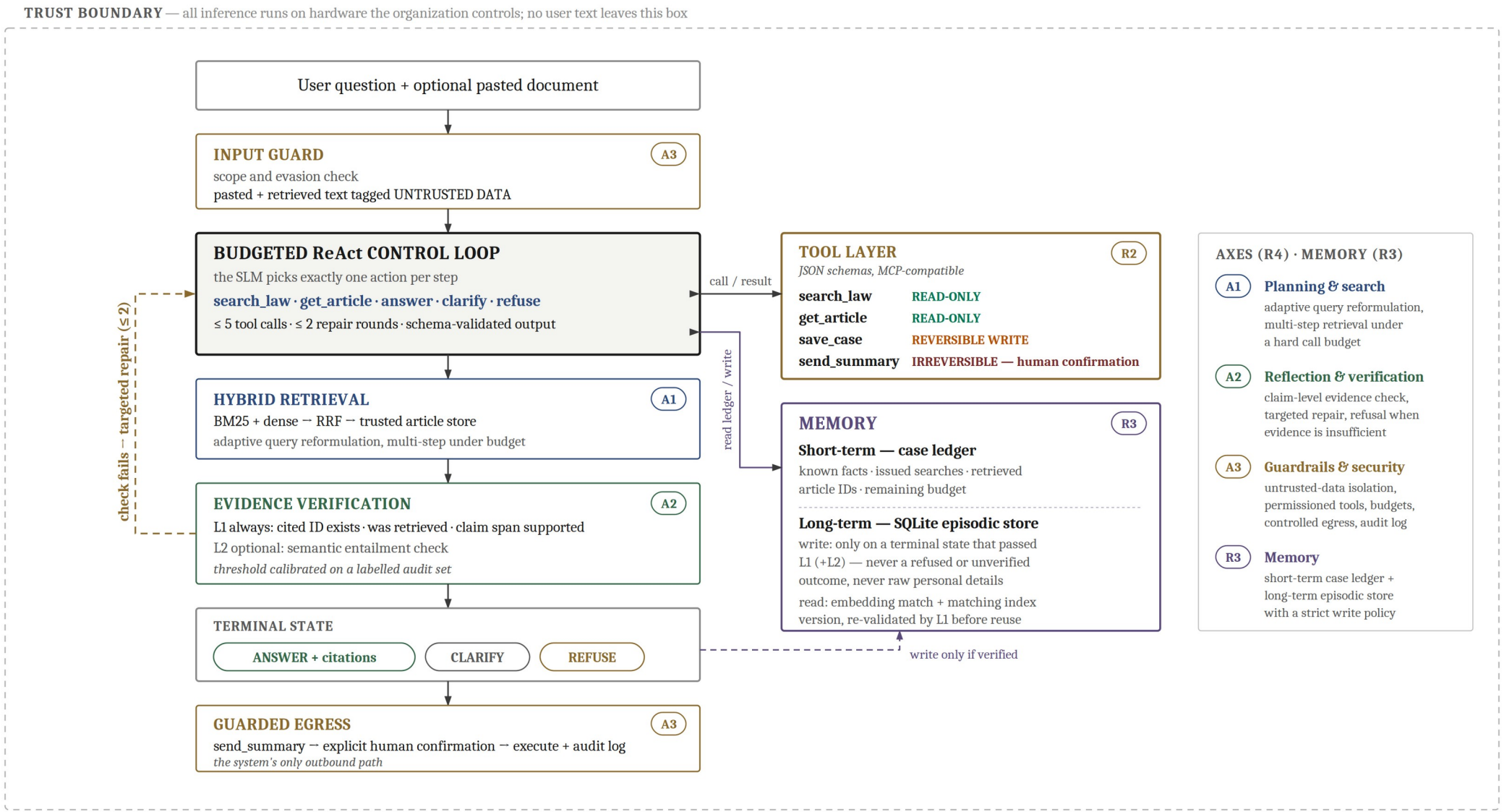
Why an agent, not a script

A script fixes its control flow before it has seen any evidence.

Answer, ask, or refuse is a function of evidence that exists only after retrieval and verification.

Retrieve-once-then-generate must commit to answering. That commitment is the failure we target.

System architecture



Evaluation plan

1:30 – 2:15

Task sets

- 200 held-out VLQA query-to-article pairs
- 20 end-to-end tasks — 10 single-provision, 5 synthesis, 3 unanswerable, 2 adversarial
- ≥ 10 injection and attack cases

Metrics

- Recall@5, Recall@20, MRR
- Unsupported-sentence rate, answer coverage
- Citation precision and recall, refusal correctness
- Attack-success rate, p50/p95 latency, compute cost

Checking the checker

- 30 hand-labelled (claim, provision) pairs audit the verifier — support accuracy, false-refusal rate
- Utility and attack-success reported separately, so refusing everything scores no better
- Failures binned into 6 named categories

3 configurations × 3 seeds × 20 tasks = 180 end-to-end runs

A local SLM, verification off

B local SLM, verification on — A vs B isolates the mechanism we claim

C Qwen3-32B self-hosted, verification on — bounds what local deployability costs in quality

Seeds 42 / 123 / 2026 · mean ± variance, no claim inside one standard error · one full run with networking disabled · zero API spend, under 20 GPU-hours

Threat model v0

2:15 – 2:45

Assets	Answer correctness · user data · the audit trail · egress points
Entry points	The user question · a pasted document · retrieved article text · tool output · the send_summary payload, which is the only egress
Attacker	Can craft arbitrary queries and pasted documents, and can plant text inside content our own retriever returns. Cannot modify the trusted legal store or the code.
Mitigation	Pasted and retrieved text is tagged untrusted and can never alter control policy or tool permissions. Read-only by default; the single egress needs explicit human confirmation and is logged before it leaves.
Tested by	An attack suite of ≥ 10 cases — 4 direct in the user turn, 4 carried by a pasted document, and ≥ 2 planted in our own retrieved content via a poisoned index shard.

Riskiest assumption

2:45 – 3:00

That a 4–8B local model can stay inside a five-action, five-call loop, recover from a failed evidence check, and do it often enough to measure something.

If wrong	4B, then 8B, then the 32B configuration — schema-constrained output and hard budgets throughout. Deterministic L1 citation validation is the floor and is never removed.
Checked in W3	Whether VLQA carries article-level gold labels, before the build starts. If it does not, all 20 end-to-end tasks are hand-authored.
Cut order	reranker and semantic-entailment ablations, then verifier audit size, then the 32B comparator
Never cut	control loop · hybrid retrieval · L1 + span verification · long-term store · guardrails · 20 tasks with one ablation · security analysis

Backup

Novelty · hardest design decision · who it matters to · week plan · requirement map · references

Novelty: each existing option fails on a different axis

Existing approach	Where it fails	What LuatCheck does
General AI assistant	Data leaves the organization; citations may be unsupported	Local inference, per-claim evidence verification
Legal information portal	Authoritative, but you must know which document to open	Conversational search over the same trusted text
One-shot RAG chatbot	Retrieves once, then generates unchecked claims	Adaptive search, repair, clarification, or refusal
Commercial legal AI	Measured to still mis-support citations	Claim-level grounding that we measure and report

The combination is the contribution: local deployment, a per-claim evidence check, and an explicit refusal path. There is no tool we could simply configure to get all three. We do not propose a new foundation model or a new retriever.

Hardest design decision

What to do when the evidence check fails. Four legitimate responses, and the right one depends on why it failed.

Retry	reformulate and search again, if the miss looks like a retrieval miss
Clarify	ask for the missing fact, if the question is under-specified
Refuse	if the provision genuinely does not cover the case
Answer narrower	if only part of the claim is supported

Getting it wrong is a failure in both directions

Too eager to refuse and the product is useless — that is the false-refusal rate. Too eager to answer and we reproduce exactly the mis-citation we set out to prevent. The accept threshold on the support score decides which way we fail, and it is an empirical question about Vietnamese legal text, not something a prompt settles.

Why an AI coding assistant cannot do this in an afternoon

It can scaffold the loop in an afternoon — and it does; we say so in our AI-use statement. What it cannot do is tell us where that threshold sits, or pay the cost of measuring the trade-off between grounding, coverage and latency on a 4B model under a five-call budget. That measurement is the project.

Who is affected, and why now

If it works perfectly, who uses it

An SME with no in-house counsel, whose HR staff answer employment questions today by guessing or by pasting a contract into a public chatbot. They get an answer plus the provision it rests on, and a refusal when the corpus does not cover the case.

The failure is measured, not hypothetical

Magesh et al. (2025) assessed leading commercial AI legal-research tools and still found unsupported and mis-attributed citations. A fluent wrong answer about notice periods has employment and compliance consequences.

The privacy constraint is now law

Vietnam's Law on Personal Data Protection (91/2025/QH15) took effect 1 January 2026, with Decree 356/2025/NĐ-CP. Employment questions carry exactly the data it regulates, which is why inference stays inside the organization.

Ten weeks, three independent tracks

Week	Outcome	Owner
W3	Interface contract frozen: tool JSON schemas, retrieval API, trace and log format — from here the three tracks are independent	all
W4–W5	Corpus indexed, BM25 + dense + RRF, first Recall@k · budgeted loop with schema-validated output and L1 · UI, permissioned tool layer, audit log, SQLite	Quyền / Dũng / Phong
W6	Span-support check and targeted repair; harness end-to-end on 10 tasks · guarded send_summary, first injection cases	Dũng, Quyền / Phong
W7 (D2)	Checkpoint: live demo, retrieval metrics, ≥10 tasks, threat model v1	all
W8–W9	Full 20-task set, fault injection, degraded mode · 3 configs × 3 seeds, verifier audit	Quyền, Phong / Dũng
W10–W11	Attack suite, error analysis, latency and cost, fully offline run, one-command reproduction	all
W12	D3 and D4 tagged; Round 1 defense with live demo	all

Two of the three of us are software engineers new to agentic AI, with about four hours a week each. Only one track is agent-specific; retrieval, UI, storage and the fault harness are conventional engineering. Each member owns a component that appears in the evaluation.

Where each requirement is satisfied

Req	What it asks	Where it lives
R2	Tools / MCP with permissions	search_law, get_article read-only · save_case reversible · send_summary irreversible, human confirmation. JSON schemas kept MCP-compatible, so the layer can be exposed as an MCP server in D2 without touching the agent.
R3	Memory	Short-term case ledger (facts, searches, article IDs, budget) + long-term SQLite episodic store. Written only on a verified terminal state; never a refusal, never raw personal details.
R4	≥ 3 agentic axes	A1 planning and search · A2 reflection and verification · A3 guardrails and security — each marked on the architecture figure.
R5	Evaluation	200 retrieval pairs + 20 end-to-end tasks · 3 configurations × 3 seeds = 180 runs · verifier audit on 30 labelled pairs · offline run.
R6	Threat model	Assets, entry points and attacker model, four threat classes with a named defense each, and an attack suite of ≥10 cases with utility and attack-success reported separately.

Honest status: nothing above is built yet — D1 is the design. Two items are deliberately outside the requirement floor: exposing the tool layer as an actual MCP server is D2 work, and the reranker and the semantic-entailment layer (L2) are optional ablations we add only if capacity allows.

References

[1] V. Magesh et al. “Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools.” Journal of Empirical Legal Studies 22 (2025), 216–242.

[2] National Assembly of Vietnam. Law on Personal Data Protection, Law No. 91/2025/QH15. Passed 26 June 2025; effective 1 January 2026.

[3] Government of Vietnam. Decree No. 356/2025/NĐ-CP on personal data protection. 2025.

[4] T.-M. Nguyen et al. “VLQA: The First Comprehensive, Large, and High-Quality Vietnamese Dataset for Legal Question Answering.” arXiv:2507.19995 [cs.CL], 2025.

[5] P. Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” NeurIPS, 2020.

[6] S. Yao et al. “ReAct: Synergizing Reasoning and Acting in Language Models.” ICLR, 2023.

[7] T. Gao, H. Yen, J. Yu, D. Chen. “Enabling Large Language Models to Generate Text with Citations.” EMNLP, 2023.

[8] J. Huang et al. “Large Language Models Cannot Self-Correct Reasoning Yet.” ICLR, 2024.

[9] K. Greshake et al. “Not What You’ve Signed Up For: Compromising Real-World LLM-Integrated Applications with Indirect Prompt Injection.” AISec, 2023.

[10] E. Debenedetti et al. “AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents.” NeurIPS Datasets and Benchmarks, 2024.

[11] S. Kapoor et al. “AI Agents That Matter.” Preprint, 2024.

[12] Anthropic. “Building Effective Agents.” Technical article, 2024.